

- Eg:- $x=20;$ → Update value or status.

iv) Deletion:-

- Remove identifiers when scope ends.
- Eg:- $\{$

$\text{int } x;$
 $\}$ → x deleted after block.

v) Scope Management:-

- Enter new scope → create new table.
- Exit scope → remove table.

2) a) Linear List (Array / Linked List):-

- Elements stored sequentially.
- Search is done linearly.

Time Complexity:-

- Search → $O(n)$.
- Insert → $O(1)$ (at beginning).
- Delete → $O(n)$

Advantages:-

- Simple to implement.
- Good for small programs.

Disadvantages:-

- Very slow for large data.
- Inefficient searching.

b) Hash Table:-

- Uses hash function to map identifier → index.
- Handles collisions using chaining or probing.

Time Complexity:-

- Search → $O(1)$ (average)
- Insert → $O(1)$
- Delete → $O(1)$